PsiLogic: Chaos-Aware Active Cancellation for Adam

Implementation, Convergence Theory, and Cross-Domain Benchmarks

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Abstract

PsiLogic multiplies an Adam-style update by a bounded, tensor-specific trust factor. The factor combines changes in a normalized gradient norm with disagreement between the current gradient and previous momentum. We state the equations implemented in the current source, including preprocessing, bias correction, epsilon placement, warmup, and optional update modes. Here, "chaos" names a gradient statistic; it is not a claim about mathematical chaos or a direct measurement of training instability.

We reassess two saved H100 benchmark snapshots. The four-task reference snapshot favors PsiLogic in mean NLP perplexity and ViT accuracy. The later partial snapshot gives a different picture: NLP perplexity is 8.15 ± 0.16 for PsiLogic and 8.20 ± 0.20 for AdamW; ViT accuracy is $26.72 \pm 0.67\%$ and $28.25 \pm 0.47\%$; ResNet accuracy is $22.08 \pm 0.23\%$ and $22.14 \pm 0.52\%$. Recorded run wall time in the later snapshot is $1.099\times$, $1.024\times$, and $1.038\times$ AdamW, respectively. No later diffusion outcome is available. These are run-time ratios; a separate saved optimizer-only microbenchmark reports a fused-step ratio of $2.27\times$ AdamW. The snapshots use different software versions and selected learning rates, so their difference does not isolate kernel fusion. We report seed-paired tests with an explicit Holm correction and retain the older results separately. No new training was performed for this revision. The evidence supports modest full-run overhead in the later workloads, but does not establish a general quality advantage over AdamW or validate every historical outcome for the current implementation.

The convergence analysis retains the full implemented detector, both Adam moments, bias correction, and the actual epsilon convention. Under explicit bounded-gradient and small-step assumptions, with parameter decay and gradient preprocessing disabled, it proves a pathwise stationarity-energy bound and $O(T^{-1})$ average squared gradient for deterministic gradients. Stochastic conclusions require controlled oracle error. Counterexamples identify failures of broader claims, and ten scalar and finite-sum lemmas are kernel-checked in Lean 4. These guarantees do not certify the benchmark configurations or establish an unconditional advantage over AdamW.

1 Motivation and scope

Adam already adapts its update through running gradient moments [1]. PsiLogic asks a narrower question: can a small additional reduction of each tensor's update help when recent gradient

statistics change or its gradient opposes accumulated momentum? It keeps Adam's coordinate-wise preconditioner and adds one scalar multiplier per tensor. Tensors can receive different damping even when they share a learning rate.

This motivation does not establish better training. A gradient opposing momentum may contain useful information, and damping it may slow progress. A sudden norm increase may also be meaningful rather than harmful. We treat the detector as a heuristic rather than calling it a reliable measure of confusion or convergence.

This manuscript revises the original PsiLogic preprint [2], first submitted to arXiv on 5 July 2026. The arXiv paper is the primary scholarly reference for PsiLogic. This local revision makes the current update explicit and reports what the saved measurements support, including unfavorable outcomes. These are distinct evidence: run metadata does not identify exact training commits or serialize every resolved optimizer setting. Equations describe the audited source; tables describe named CSV snapshots. We cannot certify that an archived run executed every equation of today's implementation.

2 Relation to existing optimizers

Adam maintains first and second moments and corrects their initialization bias [1]. AdamW separates parameter decay from the gradient direction [3]. PsiLogic follows that separation: ordinary decay acts on parameters, while cancellation scales the optimizer update. Optional Lion mode uses a sign-momentum direction [4] with the same trust factor. This is a tensor-level effective learning-rate adjustment, not a curvature estimate or replacement for an external schedule.

Gradient centralization and adaptive gradient clipping are optional preprocessing operations. They change the gradients used by both optimizer and detector. Results with these options enabled measure the full configuration and cannot isolate cancellation's contribution. We make no claim of priority over all gradient-dependent learning-rate methods. Section 4 establishes conditional convergence guarantees for a precisely specified Adam configuration and states the modes and noise regimes they exclude.

3 The implemented algorithm

3.1 State and preprocessing

Let i identify a tensor in parameter group k, and n_i its number of elements. Its counter t starts at 1 and advances only when that tensor has a gradient. A skipped tensor receives no moment, detector, or decay update. Thus t need not equal the global step. Let $\eta_{k,t}$ be the trainer-supplied learning rate, λ_k the decoupled decay coefficient, and ϵ the numerical stabilizer. We suppress group indices on other settings. Parameter-shaped products, squares, square roots, and divisions are elementwise.

Let $h_{i,t}$ be the gradient received by the optimizer, after trainer-side unscaling, accumulation, or global clipping. Tensor-wise adaptive gradient clipping (AGC) with coefficient $a > 0$ gives

$$\widetilde{g}_{i,t} = h_{i,t} \min\left(1, \frac{a \max(\|\theta_{i,t}\|_2, 10^{-3})}{\max(\|h_{i,t}\|_2, 10^{-6})}\right).$$

For $a \le 0$, $\widetilde{g} = h$. Enabled gradient centralization, on tensors with more than one dimension, gives

$$g_{i,t} = \widetilde{g}_{i,t} - \text{mean}_{d=1,\ldots,D_i-1}(\widetilde{g}_{i,t}).$$

Otherwise $g = \widetilde{g}$. Reduced dimensions remain for broadcasting; the first dimension is not averaged. Moments and detector use g. Secondary decay uses $\widetilde{g}$ before centralization. AGC reads parameters before decay. State is FP32 even with `FP16/BF16` parameters. Equations specify arithmetic, not bitwise equality across devices or dtypes.

3.2 Normalized gradient statistics

The root-mean-square gradient magnitude is

$$u_{i,t} = \frac{\|g_{i,t}\|_2}{\sqrt{\max(n_i, 1)}}.$$

Initialize $A_{i,1} = u_{i,1}$ and dimensionless averages $F_{i,1} = S_{i,1} = 1$. For $t > 1$,

$$\begin{aligned} A_{i,t} &= 0.99A_{i,t-1} + 0.01u_{i,t}, \\ r_{i,t} &= u_{i,t}/(A_{i,t} + \epsilon), \\ F_{i,t} &= 0.90F_{i,t-1} + 0.10r_{i,t}, \\ S_{i,t} &= 0.99S_{i,t-1} + 0.01r_{i,t}. \end{aligned}$$

The ratio uses newly updated A. Averages advance during warmup and with cancellation disabled. They track relative magnitude, not gradient variance. Normalization approximately removes constant gradient rescaling when epsilon is negligible; it is not exact scale invariance near numerical zero.

3.3 Continuous signal and trust

With momentum before the current update denoted by $m_{i,t-1}$, set disagreement $D_{i,1} = 0$. Subsequently,

$$\begin{aligned} c_{i,t} &= \text{clip}\left(\frac{\langle m_{i,t-1}, g_{i,t}\rangle}{\|m_{i,t-1}\|_2\|g_{i,t}\|_2 + \epsilon}, -1, 1\right), \\ D_{i,t} &= \tfrac{1}{2}(1 - c_{i,t}). \end{aligned}$$

Epsilon is inside the cosine denominator. After the first update, zero momentum or zero gradient gives $D = 1/2$, not the first-step value. With $\sigma(x) = 1/(1 + e^{-x})$, adaptive scale τ_s and fixed setting τ_c, define

$$E_{i,t} = \begin{cases} \sigma\left(\dfrac{4}{\max(\tau_s - 1, 0.25)}\left(\dfrac{F_{i,t}}{S_{i,t} + \epsilon} - 1\right)\right), & \text{adaptive}, \\ \sigma\left(4\left(\dfrac{S_{i,t}}{\max(\tau_c, \epsilon)} - 1\right)\right), & \text{fixed}. \end{cases}$$

Fixed mode also uses normalized slow state, not a raw-norm threshold. When the cancellation path is active,

$$C_{i,t} = \tanh(S_{i,t})\left(\tfrac{1}{2}E_{i,t} + \tfrac{1}{2}D_{i,t}\right).$$

With warmup gain w, effective gamma γ, amplification P_k, and $b = \min(b_0, 0.99)$,

$$T_{i,t} = 1 - \min(C_{i,t}\gamma_{i,t}P_k w_{i,t}, b).$$

For intended nonnegative amplification, $1 - b \le T \le 1$. Default $b_0 = 0.05$ gives $T \ge 0.95$. The code clamps cancellation only from above and does not reject negative amplification; these bounds assume $P_k \ge 0$.

If gain or effective gamma is zero, the path is inactive: trust is 1 and the stored soft signal is zero. This is an inactive marker, not a measurement of quiet gradients. Equal fast and slow state does not close the gate: their ratio near 1 gives $E \approx 1/2$. With $S \approx 1$ and aligned gradients, $C \approx \tanh(1)/4 \approx 0.1904$. Gamma 0.05, amplification 1, and full gain then yield about 0.95% damping. Stable directions and small absolute gradients do not guarantee zero cancellation; a limit such as $S \to 0$ does force the active signal to zero.

3.4 Schedules and warmup

For configured horizon H, the common gamma and secondary-decay schedule is

$$z_t = \begin{cases} \frac{1}{2}[1 + \cos(\pi \min(t/H, 1))], & H > 0, \\ 1, & H = 0. \end{cases}$$

Set $\gamma_t^{\text{sched}} = \gamma_0 z_t$ and $q_t = q_0 z_t$. Optional automatic gamma multiplies scheduled gamma by $\max(S_{i,t}/0.1, 0.1)$ when $t > 1$ and $S_{i,t} < 0.1$, and leaves it unchanged otherwise. This option is off by default and is not a convergence test.

Automatic chaos warmup gives $W = \max(500, \lfloor H/20 \rfloor)$ for $H > 0$, or 200 for $H = 0$. An explicitly supplied nonnegative length overrides it. With $R = \max(1, \lfloor W/4 \rfloor)$,

$$w_{i,t} = \begin{cases} 0, & t \le W, \\ \min(1, (t - W)/R), & t > W. \end{cases}$$

Warmup zero activates the path on the first update. The bare constructor with unknown horizon waits 200 updates, then ramps for 50. This is separate from trainer LR warmup. Each tensor's counter drives both internal schedules.

3.5 Adam update, epsilon, and decay

Starting with $m_{i,0} = v_{i,0} = 0$,

$$m_{i,t} = \beta_1 m_{i,t-1} + (1 - \beta_1) g_{i,t},$$
$$v_{i,t} = \beta_2 v_{i,t-1} + (1 - \beta_2) g_{i,t}^{\odot 2}.$$

Ordinary decay multiplies parameters by $1 - \eta_{k,t}\lambda_k$. If $q_t > 0$ and cancellation is active ($w > 0$, $\gamma > 0$), secondary decay adds

$$\rho_{i,t} = 1 - \eta_{k,t} q_t w_{i,t} (1 - C_{i,t}) \tanh(|\widetilde{g}_{i,t}|).$$

Otherwise $\rho = 1$. The API calls this option quantum_decay; it has no quantum-mechanical interpretation and is disabled by default. Both decay factors act before the optimizer direction:

$$\boxed{\theta_{i,t+1} = \rho_{i,t}(1 - \eta_{k,t}\lambda_k)\theta_{i,t} - \eta_{k,t} \frac{\sqrt{1 - \beta_2^t}}{1 - \beta_1^t} T_{i,t} \frac{m_{i,t}}{\sqrt{v_{i,t}} + \epsilon}.}$$

With bias-corrected moments $\widehat{m} = m/(1-\beta_1^t)$ and $\widehat{v} = v/(1-\beta_2^t)$, the direction is

$$\frac{T_{i,t}\widehat{m}_{i,t}}{\sqrt{\widehat{v}_{i,t}} + \epsilon/\sqrt{1-\beta_2^t}}.$$

Replacing that denominator with $\sqrt{\widehat{v}} + \epsilon$ changes the algorithm. For $\beta_2 = 0.999$, first-update effective epsilon is about 31.62ϵ. With all extras disabled, this is AdamW-style arithmetic with that convention, not exact algebraic identity to usual PyTorch AdamW. Tolerance-based equivalence checks do not prove exact equality.

3.6 Lion and distributed execution

Optional Lion mode gives

$$\begin{aligned} d_{i,t} &= \operatorname{sign}(\beta_1 m_{i,t-1} + (1-\beta_1) g_{i,t}), \\ \theta_{i,t+1} &= \rho_{i,t}(1 - \eta_{k,t}\lambda_k)\theta_{i,t} - \eta_{k,t} T_{i,t} d_{i,t}, \\ m_{i,t} &= \beta_2 m_{i,t-1} + (1-\beta_2) g_{i,t}. \end{aligned}$$

Disagreement still uses previous momentum. Lion has no Adam bias correction. The shared initializer allocates a second-moment buffer, but it does not enter this update.

Optional distributed detector synchronization averages F, S across ranks after their local update and before computing the signal. It does not explicitly synchronize A or disagreement. Averaging F, S cannot guarantee identical trust if ranks differ in gradients, momentum, counters, or settings. Ordinary DDP with matching reduced gradients and state is the intended setting.

3.7 Execution order, defaults, and source map

1. Skip tensors without gradients; apply optional AGC, then centralization.
2. Increment active tensor counters; when cancellation can activate, compute disagreement against previous momentum.
3. Update Adam moments in Adam mode and norm averages in both modes.
4. Optionally average `fast/slow` state across distributed ranks.
5. Resolve effective gamma, signal, and trust. Apply ordinary decay, then active secondary decay.
6. Apply the direction; update Lion momentum after constructing its direction.

Bare defaults are LR 0.001, betas (0.9, 0.999), decay 0.0001, epsilon 1e-8, gamma 0.05, amplification 1, cap 0.05, adaptive thresholding, tau scale 2, and automatic chaos warmup. AGC, centralization, secondary decay, automatic gamma, Lion, and detector synchronization are off. CUDA foreach and fused execution are requested by default with availability-dependent fallback. CPU and compilation use the scalar implementation. Presets and parameter groups can override these settings.

API mapping: A, F, S are gn_avg, fast, slow; C is soft_chaos; P_k is p_ext; b_0 is max_cancel; τ_s, τ_c are tau_scale, chaos_tau; H is gamma_T_max; q_0 is quantum_decay; a is agc_clip. Preprocessing and update order are in `psilogic/optimizer.py`; detector helpers in `psilogic/_chaos.py`; accelerated execution in `psilogic/_cuda/step.py` and kernels.py. Backends target the same rule, with possible precision and reduction-order differences. Software checks cannot establish which source a historical run executed.

4 Convergence Theory and Proofs

This section supplies the complete conditional convergence analysis for the update described in Section 3. It is placed before the empirical protocol to make the mathematical guarantee and its limits explicit before the benchmark results are interpreted. The deterministic guarantee, stochastic error bounds, and counterexamples concern precisely stated configurations; they do not validate every benchmark preset.

For the proofs, x_t concatenates the parameter tensors $\theta_{i,t}$, $u_t^{(b)}$ denotes the trust $T_{i,t}$, and s_t denotes the soft signal $C_{i,t}$ of the implementation section. The notation $h_t = \nabla f(x_t)$ means the true objective gradient here; the earlier implementation notation $h_{i,t}$ meant the received raw gradient. This distinction is needed to state oracle error. The recurrence below restates the same algorithm in this proof notation, preserving its epsilon convention and update order. All sidecar filenames and the `lean/` directory named in this section are relative to `research/convergence/`.

4.1 Scope and mathematical contributions

Adaptive normalization and momentum are useful algorithmic mechanisms, but neither positive step multipliers nor small empirical loss establish a convergence theorem. PsiLogic adds a multiplier computed from the same gradient used in its update. This creates a conditional dependence that must be retained in the analysis. The present object of study is the local package at revision `b4917163d63d8a5cb2960e29af4b1bccdbf5798d`, with its working-tree state recorded in `source_snapshot.json`. Existing papers, documentation, and release equations were not used as the algorithm specification.

The main contributions are: (i) exact equations and branch coverage; (ii) uniform real-arithmetic bounds on the implemented detector; (iii) a pathwise error-energy theorem for bias-corrected momentum with arbitrary, possibly current-gradient-dependent, uniformly positive diagonal scaling; (iv) its specialization to an actual PsiLogic configuration, with explicit constants; and (v) counterexamples defining the limits of stronger claims. The main theorem does not assume monotonic second moments, independence of trust and gradients, or the convergence of classical Adam. Its sufficient step bound can be extremely conservative when epsilon is small. It is a rigorous existence guarantee, not a recommendation for training hyperparameters.

4.2 Notation and recurrence for the proofs

4.2.1 Blocks, clocks, defaults, and preprocessing

Partition $x \in \mathbb{R}^d$ into parameter tensors $x^{(b)} \in \mathbb{R}^{d_b}$; tensor shape is retained for centralization. A block with `grad is None` receives no update, no decay, and no local-clock increment. In this section $k \geq 1$ is a block's local active-update count; x denotes its pre-step value. Group options may differ. Unless otherwise specified, all formulas are exact-real interpretations, not statements about floating-point evaluation.

The constructor defaults are

$$\eta = 10^{-3}, \quad (\beta_1, \beta_2) = (0.9, 0.999), \quad \lambda = 10^{-4}, \quad \gamma = 0.05, \quad p_{\text{ext}} = 1,$$
$$q = 0, \quad \epsilon = 10^{-8}, \quad c_{\max} = 0.05, \quad \tau = 0.5, \quad \tau_{\text{scale}} = 2, \quad H = 0, \quad W_{\text{cfg}} = -1.$$

AGC, centralization, Lion, auto-gamma, and DDP chaos synchronization default to false (AGC coefficient zero); adaptive thresholding, foreach, and fused CUDA selection default to true. Task presets and parameter-group builders can change these choices. No global norm clipping is implemented by this optimizer; external training-loop clipping is outside its update path.

Given raw supplied gradient z, AGC with coefficient $a_{\mathrm{gc}} > 0$ computes

$$\chi = \min\left\{1, \frac{a_{\mathrm{gc}} \max(\|x\|, 10^{-3})}{\max(\|z\|, 10^{-6})}\right\}, \qquad r = \chi z. \tag{4.1}$$

With AGC disabled, $r = z$. Centralization gives $g = Pr$, where P subtracts the mean over every axis except the leading axis if tensor rank exceeds one; otherwise $P = I$. All moments and detector norms use g. Quantum decay uses r, specifically *after AGC and before centralization*. The norm in (4.1) is a whole-tensor Euclidean norm, not a per-row norm.

4.2.2 Detector, schedules, and cancellation

The initialized state is $m_0 = v_0 = 0$, $A_0 = F_0 = S_0 = 0$, stored soft chaos zero, and local count zero. Moments are block vectors; A, F, S and soft chaos are one-element tensors. Define $R_k = \|g_k\| / \sqrt{\max(d_b, 1)}$. At the first active step,

$$A_1 = R_1, \qquad F_1 = S_1 = 1, \tag{4.2}$$

even when $g_1 = 0$. At later steps,

$$A_k = .99A_{k-1} + .01R_k, \quad Z_k = \frac{R_k}{A_k + \epsilon}, \quad F_k = .9F_{k-1} + .1Z_k, \quad S_k = .99S_{k-1} + .01Z_k. \tag{4.3}$$

If enabled and a process group exists, DDP replaces F_k, S_k by their arithmetic means across ranks. It does not average A_k, local disagreement, or local moment state. Consistent collective ordering is required.

Disagreement reads the *old* momentum:

$$D_k = \begin{cases} 0, & k = 1, \\ \frac{1}{2}\left[1 - \mathrm{clip}\left(\dfrac{\langle g_k, m_{k-1}\rangle}{\|g_k\|\,\|m_{k-1}\| + \epsilon}, -1, 1\right)\right], & k > 1. \end{cases} \tag{4.4}$$

It is computed only when cancellation can be active. Write $\sigma(u) = 1/(1 + e^{-u})$. The excess and soft chaos are

$$E_k = \begin{cases} \sigma\left(\dfrac{4}{\max(\tau_{\mathrm{scale}} - 1, .25)}\left(\dfrac{F_k}{S_k + \epsilon} - 1\right)\right), & \text{adaptive}, \\ \sigma\left(4\left(\dfrac{S_k}{\max(\tau, \epsilon)} - 1\right)\right), & \text{otherwise}, \end{cases} \qquad s_k = \tanh(S_k)\frac{E_k + D_k}{2}. \tag{4.5}$$

The detector is continuous on its positive-denominator domain, including clamping and the sigmoid; it has no hard spike indicator. The first-step seed and schedule branches are discrete decisions.

For horizon $H =$ `gamma_T_max` set

$$C_k = \begin{cases} (1 + \cos(\pi \min(k/H, 1)))/2, & H > 0, \\ 1, & H \le 0, \end{cases} \qquad \gamma_k^{\mathrm{sched}} = \gamma C_k, \quad q_k^{\mathrm{sched}} = qC_k.$$

For nonnegative configured warmup W_{cfg} use $W = W_{\mathrm{cfg}}$; otherwise $W = \max(500, \lfloor H/20 \rfloor)$ if $H > 0$, and $W = 200$ otherwise. Then

$$w_k = \begin{cases} 0, & k \le W, \\ \min(1, (k - W)/\max(1, \lfloor W/4 \rfloor)), & k > W. \end{cases}$$

Auto-gamma, if enabled, multiplies $\gamma_k^{\mathrm{sched}}$ by $\max(S_k/.1, .1)$ when $k > 1$ and $S_k < .1$, and by one otherwise. Denote the result by γ_k^{eff}. This uses the current, possibly synchronized, slow state. In the active branch $w_k > 0$, $\gamma_k^{\mathrm{eff}} > 0$, set

$$c_* = \min(c_{\max}, .99), \qquad u_k = 1 - \min\{s_k \gamma_k^{\mathrm{eff}} p_{\mathrm{ext}} w_k, c_*\}, \qquad Q_k = q_k^{\mathrm{sched}} w_k (1 - s_k). \tag{4.6}$$

Here Q_k is used only if $q_k^{\mathrm{sched}} > 0$; otherwise $Q_k = 0$. In an inactive branch $u_k = 1, Q_k = 0$, and the stored soft chaos is set to zero. Thus $\gamma = 0$ also disables quantum decay, even when $q > 0$.

4.2.3 Adam and Lion are different recurrences

In Adam mode,

$$m_k = \beta_1 m_{k-1} + (1 - \beta_1) g_k, \qquad v_k = \beta_2 v_{k-1} + (1 - \beta_2) g_k^{\odot 2}, \tag{4.7}$$

$$x^+ = (1 - \eta_k \lambda)(1 - \eta_k Q_k \tanh |r_k|) \odot x - \eta_k \frac{\sqrt{1 - \beta_2^k}}{1 - \beta_1^k} \frac{u_k m_k}{\sqrt{v_k} + \epsilon}. \tag{4.8}$$

All vector products, square roots, and quotients in (4.8) are coordinatewise. Weight decay precedes quantum decay; their product contains a cross term and neither is scaled by trust. With $\widehat{m}_k = m_k/(1 - \beta_1^k)$ and $\widehat{v}_k = v_k/(1 - \beta_2^k)$, the adaptive term is

$$-\eta_k u_k \frac{\widehat{m}_k}{\sqrt{\widehat{v}_k} + \epsilon/\sqrt{1 - \beta_2^k}}. \tag{4.9}$$

This epsilon placement differs from the commonly written Adam update with constant epsilon in the bias-corrected denominator.

In Lion mode the recurrence is instead

$$\begin{aligned} \ell_k &= \operatorname{sign}\big(\beta_1 m_{k-1} + (1 - \beta_1) g_k\big), \\ x^+ &= (1 - \eta_k \lambda)(1 - \eta_k Q_k \tanh |r_k|) \odot x - \eta_k u_k \ell_k, \\ m_k &= \beta_2 m_{k-1} + (1 - \beta_2) g_k, \end{aligned} \tag{4.10}$$

with $\operatorname{sign}(0) = 0$, no bias correction, and unchanged variance state. Our main convergence theorem concerns (4.8), not (4.10).

4.3 Problem formulation and explicit assumptions

For a positive integer horizon T, use global indices $t = 1, \ldots, T$, with x_t the point before update t and $h_t = \nabla f(x_t)$. Let $\mathcal{F}_{t-1}$ contain x_1, all gradients observed before update t, pre-step optimizer states, and any history-dependent group settings. Thus $x_t, h_t, m_{t-1}, v_{t-1}, A_{t-1}, F_{t-1}, S_{t-1}$ are $\mathcal{F}_{t-1}$-measurable. With the current gradient included they generate $\mathcal{F}_t$. Preprocessed gradients, disagreement, updated detector states, trust, updated moments, and x_{t+1} are $\mathcal{F}_t$-measurable, generally not $\mathcal{F}_{t-1}$-measurable. In a distributed formulation the filtration must include the ranks' joint observations and completed collectives.

Assumption 4.1 (Analytic domain)**.** $f : \mathbb{R}^d \to \mathbb{R}$ has L-Lipschitz gradient, $L > 0$, and $f \geq f_* > -\infty$. All iterates and states are finite real quantities. Every relevant group has $\epsilon > 0$, $0 \leq \beta_1, \beta_2 < 1$, finite $\gamma, q \geq 0$, finite $p_{\mathrm{ext}} \geq 0$, and $0 < c_{\max} \leq 1$. Local clocks are nonnegative integers; warmup and horizon settings are fixed finite integers. Initial moments and detector states are as in Section 4.2. There are no external parameter mutations.

These conditions are stronger than constructor acceptance. Epsilon and p_{ext} are not checked there, and arbitrary group overrides bypass some top-level checks. They are mathematical scope restrictions on supported inputs, not changes to executable code.

Assumption 4.2 (Canonical configuration for the main theorem)**.** Use Adam mode with AGC and centralization disabled, $\lambda = q = 0$, and all blocks active at every step. Groups share a constant learning rate $\alpha > 0$, fixed $\beta_1 = \beta$, fixed β_2, and a common $\epsilon > 0$. Detector settings may differ by block within Assumption 4.1; the exact implemented detector, warmup, cosine schedule, and auto-gamma are retained. Use ordinary initialized state, not an arbitrary resumed checkpoint. The supplied gradients satisfy $\|g_t\| \le G$ for a finite deterministic G. There is no distributed execution requirement: the theorem is stated for a single process and the real recurrence of the scalar path.

The default positive weight decay is deliberately excluded. The main theorem is an actual selectable configuration of PsiLogic, not a claim about every constructor default. All proofs are for the real recurrence; fp32 state is not treated as exact real arithmetic. Equality of intended backend equations is separately qualified by tensor layout and precision.

Definition 4.3 (Oracle error and gap)**.** Set $\xi_t = g_t - h_t$, $\Delta = f(x_1) - f_*$, and $\Xi_T = \sum_{t=1}^{T} \|\xi_t\|^2$. The main pathwise theorem does not require unbiasedness or stochastic independence. Stochastic corollaries assume the stated second moments exist; x_1 is deterministic unless Δ is replaced by its expectation.

4.4 Preliminary bounds for the exact detector

Lemma 4.4 (Normalized EMA stability)**.** *For a nonempty block satisfying Assumption 4.1, $A_k \ge 0$ and $0 \le Z_k \le 100$ for $k \ge 2$. In the absence of DDP, $0 \le F_k, S_k \le 100$. These bounds also survive averaging nonnegative bounded rank states. If the postprocessed block gradient tends to zero and DDP is disabled, then $A_k, F_k, S_k \to 0$.*

Proof. The first seed has $A_1 = R_1 \ge 0$ and $F_1 = S_1 = 1$. Positivity of the EMA coefficients proves $A_k \ge 0$ by induction. For $k \ge 2$, $A_k + \epsilon \ge .01R_k + \epsilon > 0$, so $0 \le R_k/(A_k + \epsilon) \le 100$. The updates of F, S are convex combinations of their old values and Z_k, so induction gives the interval bound. Arithmetic means preserve that interval.

For completeness, if $b \in [0, 1)$ and $y_k = by_{k-1} + (1 - b)a_k$ with $a_k \to 0$, then $y_k = b^{k-n}y_n + (1 - b)\sum_{j=n+1}^{k} b^{k-j}a_j$. Given $\delta > 0$, choose n so $|a_j| \le \delta$ for $j > n$; the sum is at most δ in magnitude and the first term tends to zero. Thus $y_k \to 0$. Apply this argument to A when $R_k \to 0$. Since $Z_k \le R_k/\epsilon$, $Z_k \to 0$; applying it again to F, S proves the claim. □

Lemma 4.5 (Trust and effective-time bounds)**.** *Under Assumption 4.1, let $c = \min_b[1 - \min(c_{\max}^{(b)}, .99)]$. Then $c \ge .01$ and each active or inactive trust satisfies $c \le u_t^{(b)} \le 1$. For any nonnegative learning rates,*

$$c\sum_{t=1}^{T} \eta_t \le \sum_{t=1}^{T} \eta_t u_t^{(b)} \le \sum_{t=1}^{T} \eta_t.$$

The detector cannot by itself turn a nonsummable learning-rate sequence into a summable one.

Proof. The clipped cosine lies in $[-1, 1]$, so $0 \le D \le 1$. The sigmoid lies in $(0, 1)$ and $0 \le \tanh S \le 1$; hence $0 \le s \le 1$. The schedule and auto-gamma are nonnegative, and $w \in [0, 1]$. Consequently the argument of the minimum in (4.6) is nonnegative, and the subtracted amount lies in $[0, c_*]$. Inactive trust is one. Multiplying $c \le u_t^{(b)} \le 1$ by $\eta_t \ge 0$ and summing proves the assertion. This says nothing about normalization, momentum direction, or stochastic bias. □

Lemma 4.6 (Bounds for the exact Adam scaling)**.** *Under Assumptions 4.1–4.2, write*

$$x_{t+1} = x_t - \alpha H_t \widehat{m}_t, \qquad (H_t)_{ii} = \frac{u_t^{(b(i))}\sqrt{1-\beta_2^t}}{\sqrt{v_{t,i}}+\epsilon}. \tag{4.11}$$

Then, in the Loewner order,

$$aI \preceq H_t \preceq MI, \qquad a = \frac{c}{G + \epsilon/\sqrt{1-\beta_2}} > 0, \qquad M = \epsilon^{-1}. \tag{4.12}$$

Also $\|\widehat{m}_t\| \le G$.

Proof. Expanding the variance recurrence gives $v_{t,i} = (1-\beta_2)\sum_{j=1}^t \beta_2^{t-j} g_{j,i}^2 \le G^2(1-\beta_2^t) \le G^2$, and it is nonnegative. Dividing numerator and denominator in (4.11) by $\sqrt{1-\beta_2^t}$ gives a denominator at most $G + \epsilon/\sqrt{1-\beta_2}$ and numerator at least c. This proves the lower bound. In the original form the numerator is at most one and the denominator at least ϵ, proving the upper bound; in particular $a \le M$. Expansion of the first moment gives

$$\widehat{m}_t = \sum_{j=1}^t w_{tj} g_j, \qquad w_{tj} = \frac{(1-\beta)\beta^{t-j}}{1-\beta^t}, \qquad w_{tj} \ge 0, \quad \sum_{j=1}^t w_{tj} = 1. \tag{4.13}$$

The triangle inequality gives the moment bound. For $\beta = 0$, these weights are understood as $w_{tt} = 1$ and $w_{tj} = 0$ for $j < t$. □

4.5 A simplified stochastic baseline

This section replaces momentum and adaptive variance by a predictable symmetric scaling B_t. For example, a scalar detector evaluated using only past gradients could supply such a multiplier. This is a *modified algorithm*; using the current implemented u_t does not satisfy its predictability assumption.

Theorem 4.7 (Predictable scaling)**.** *Suppose* $x_{t+1} = x_t - \alpha B_t g_t$*, with fixed constants* $0 < a \le M$*,* $aI \preceq B_t \preceq MI$*,* B_t *measurable with respect to* $\mathcal{F}_{t-1}$*,* $\mathbb{E}[g_t \mid \mathcal{F}_{t-1}] = h_t$*, and* $\mathbb{E}[\|g_t - h_t\|^2 \mid \mathcal{F}_{t-1}] \le \sigma^2$*. If* f *obeys Assumption 4.1 and* $0 < \alpha \le a/(LM^2)$*, then*

$$\frac{1}{T}\sum_{t=1}^T \mathbb{E}\,\|h_t\|^2 \le \frac{2\Delta}{a\alpha T} + \frac{LM^2\alpha\sigma^2}{a}. \tag{4.14}$$

Choosing $\alpha = \min\{a/(LM^2), T^{-1/2}\}$ *gives* $O(T^{-1/2})$ *for fixed problem constants.*

Proof. Lipschitz continuity of the gradient implies the smoothness inequality $f(x+s) \le f(x) + \langle \nabla f(x), s\rangle + (L/2)\|s\|^2$: indeed integrate $\langle \nabla f(x+rs) - \nabla f(x), s\rangle$ from $r = 0$ to 1 and bound it by $\int_0^1 Lr\|s\|^2\,dr$. Apply it with $s = -\alpha B_t g_t$. These random variables are integrable at every finite time: from deterministic x_1, the bound $\|h_t\| \le \|\nabla f(0)\| + L\|x_t\|$ and the conditional variance hypothesis imply finite second moment of g_t whenever x_t has finite second moment. The recursion and $\|B_t\| \le M$ then imply finite second moment of x_{t+1}. Induction proves these facts, and $|f(x)| \le |f(0)| + \|\nabla f(0)\|\,\|x\| + (L/2)\|x\|^2$ implies objective integrability. Thus the conditional expectations below are defined. Predictability and conditional unbiasedness give $\mathbb{E}[\langle h_t, B_t g_t\rangle \mid \mathcal{F}_{t-1}] = \langle h_t, B_t h_t\rangle \ge a\|h_t\|^2$. Expansion of $g_t = h_t + (g_t - h_t)$ gives conditional second moment at most $\|h_t\|^2 + \sigma^2$, with zero cross term. Thus

$$\mathbb{E}[f(x_{t+1}) \mid \mathcal{F}_{t-1}] \le f(x_t) - \alpha(a - L\alpha M^2/2)\|h_t\|^2 + L\alpha^2 M^2\sigma^2/2.$$

The step condition leaves coefficient at least $a/2$. Take expectations, sum, and use $\mathbb{E}f(x_{T+1}) \ge f_*$. Division by $a\alpha T/2$ yields (4.14). The specified horizon-dependent step has the stated asymptotic order. No factorization involving a current-gradient scaling was used. □

4.6 Exact cancellation, moments, and bias correction

4.6.1 A pathwise momentum-tracking lemma

The following argument permits H_t to depend arbitrarily on the current gradient. This is the key distinction from Theorem 4.7. Define

$$K = \frac{\beta}{1-\beta}, \qquad C_\beta = \frac{1}{1-\beta}, \qquad e_t = \widehat{m}_t - h_t.$$

Lemma 4.8 (Finite-horizon tracking energy)**.** *For $g_t = h_t + \xi_t$, the weights in (4.13), and any sequence $h_1, \dots, h_T$, let $n_t = \sum_{j=1}^t w_{tj}\xi_j$ and $d_t = \sum_{j=1}^t w_{tj}h_j - h_t$. Then $e_t = n_t + d_t$ and*

$$\sum_{t=1}^T \|n_t\|^2 \le C_\beta \sum_{t=1}^T \|\xi_t\|^2, \tag{4.15}$$

$$\sum_{t=1}^T \|d_t\|^2 \le K^2 \sum_{t=2}^T \|h_t - h_{t-1}\|^2. \tag{4.16}$$

For the update (4.11) and an L-Lipschitz gradient, therefore,

$$\sum_{t=1}^T \|e_t\|^2 \le 2C_\beta \Xi_T + 2Q \sum_{t=1}^T \|\widehat{m}_t\|^2, \qquad Q = K^2L^2\alpha^2M^2. \tag{4.17}$$

Proof. The identity follows by expanding $g_j = h_j + \xi_j$. Convexity of squared norm (or Cauchy–Schwarz with weights summing to one) gives $\|n_t\|^2 \le \sum_{j\le t} w_{tj}\|\xi_j\|^2$. Since $1-\beta^t \ge 1-\beta$, $w_{tj} \le \beta^{t-j}$; hence $\sum_{t=j}^T w_{tj} \le \sum_{r=0}^\infty \beta^r = C_\beta$. Interchanging the two *finite* sums proves (4.15).

Let $b_t = \sum_{j\le t} w_{tj}h_j$. For $t \ge 2$, $b_t = \rho_t b_{t-1} + (1-\rho_t)h_t$, where

$$\rho_t = \frac{\beta(1-\beta^{t-1})}{1-\beta^t} \in [0, \beta].$$

Thus $d_1 = 0$ and $d_t = \rho_t(d_{t-1} + h_{t-1} - h_t)$. Iterating its norm inequality gives $\|d_t\| \le \sum_{j=2}^t \beta^{t-j+1}\|h_{j-1} - h_j\|$. For $\beta > 0$, weighted Cauchy–Schwarz bounds its square by

$$\left(\sum_{j=2}^t \beta^{t-j+1}\right)\left(\sum_{j=2}^t \beta^{t-j+1}\|h_{j-1} - h_j\|^2\right).$$

The first factor is at most K; summing the second factors over t gives at most $K\sum_{j=2}^T \|h_{j-1} - h_j\|^2$. This proves (4.16). If $\beta = 0$, $d_t = 0$ and the result holds directly. Finally $\|n_t + d_t\|^2 \le 2\|n_t\|^2 + 2\|d_t\|^2$, while

$$\|h_t - h_{t-1}\| \le L\|x_t - x_{t-1}\| \le L\alpha M\|\widehat{m}_{t-1}\|.$$

The last sum ranges over $t-1 = 1, \dots, T-1$ and can be enlarged to $1, \dots, T$, since all terms are nonnegative. This proves (4.17). The zero initial drift and the first oracle error $n_1 = \xi_1$ have both been retained. □

4.6.2 Main theorem and proof

Theorem 4.9 (Canonical Adam configuration: pathwise bound)**.** *Under Assumptions 4.1–4.2, use a, M from (4.12), K, C_β above, and the constant step*

$$0 < \alpha \le \frac{a}{4LM^2\max\{1, K\}}. \tag{4.18}$$

Then every finite trajectory of the exact real recurrence satisfies

$$\sum_{t=1}^{T} \|\nabla f(x_t)\|^2 \le \frac{9\Delta}{\alpha a} + 13C_\beta \frac{M^2}{a^2} \sum_{t=1}^{T} \|g_t - \nabla f(x_t)\|^2 . \tag{4.19}$$

The result permits all dependencies introduced by the exact detector and second-moment normalization. It assumes no independence, no monotonicity of H_t, and no relation such as $\beta_1^2 < \beta_2$.

Proof. Write $z_t = \widehat{m}_t$ for this proof, so $h_t = z_t - e_t$. Using $aI \preceq H_t \preceq MI$, Cauchy–Schwarz, and $M\|e_t\|\,\|z_t\| \le (a/2)\|z_t\|^2 + (M^2/(2a))\|e_t\|^2$,

$$\langle h_t, H_t z_t\rangle = \langle z_t, H_t z_t\rangle - \langle e_t, H_t z_t\rangle \ge \frac{a}{2}\|z_t\|^2 - \frac{M^2}{2a}\|e_t\|^2 .$$

The smoothness inequality proved in Theorem 4.7 gives

$$f(x_{t+1}) \le f(x_t) - \alpha\left(\frac{a}{2} - \frac{L\alpha M^2}{2}\right)\|z_t\|^2 + \frac{\alpha M^2}{2a}\|e_t\|^2 . \tag{4.20}$$

Let $U = \sum_{t=1}^{T}\|z_t\|^2$. Summing (4.20), using $f(x_{T+1}) \ge f_*$ and (4.17), yields

$$\alpha\left(\frac{a}{2} - \frac{L\alpha M^2}{2} - \frac{M^2 Q}{a}\right) U \le \Delta + \frac{\alpha M^2 C_\beta}{a}\Xi_T. \tag{4.21}$$

Condition (4.18) implies $L\alpha M^2/2 \le a/8$ and

$$Q = K^2L^2\alpha^2M^2 \le \frac{a^2}{16M^2} \le \frac{1}{16}.$$

Consequently $M^2Q/a \le a/16$, so the coefficient in parentheses is at least $5a/16 \ge a/4$. Therefore

$$U \le \frac{4\Delta}{\alpha a} + \frac{4M^2C_\beta}{a^2}\Xi_T. \tag{4.22}$$

Because $h_t = z_t - e_t$,

$$\sum_{t=1}^{T}\|h_t\|^2 \le 2U + 2\sum_{t}\|e_t\|^2 \le (2+4Q)U + 4C_\beta\Xi_T \le \frac{9}{4}U + 4C_\beta\Xi_T.$$

Substitution of (4.22) gives coefficient 9 on $\Delta/(\alpha a)$ and coefficient $C_\beta(9M^2/a^2+4) \le 13C_\beta M^2/a^2$ on Ξ_T, since $a \le M$. This proves (4.19). Every inequality was pathwise before any expectation was taken. □

Corollary 4.10 (Deterministic stationarity)**.** *If $g_t = \nabla f(x_t)$ in Theorem 4.9, then*

$$\frac{1}{T}\sum_{t=1}^{T}\|\nabla f(x_t)\|^2 \le \frac{9\Delta}{\alpha a T}.$$

For an infinite trajectory satisfying its assumptions, $\|\nabla f(x_t)\| \to 0$. This does not assert that x_t converges or that its limit, if any, is a local minimum.

Proof. Set $\Xi_T = 0$. The partial sums of nonnegative squared gradients are bounded uniformly in T; hence their series converges. If its terms did not tend to zero, infinitely many would exceed one fixed positive number, contradicting boundedness of the partial sums. □

Corollary 4.11 (Stochastic error and growing batches)**.** *If the pathwise assumptions of Theorem 4.9 hold almost surely and* $\mathbb{E}\,\|\xi_t\|^2 \le \sigma_t^2$*, then*

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\,\|h_t\|^2 \le \frac{9\Delta}{\alpha a T} + 13C_\beta\frac{M^2}{a^2}\frac{1}{T}\sum_{t=1}^{T}\sigma_t^2. \tag{4.23}$$

If $\sum_t \sigma_t^2 < \infty$*, the gradients tend to zero both in mean square and almost surely. If only their Cesàro mean tends to zero, the displayed average tends to zero; this alone is not a last-iterate assertion.*

In particular, suppose each g_t *is the average of* B *conditionally independent unbiased samples, each with conditional variance at most* σ^2*, and the averages satisfy the required bound* G*. For a run of* T *iterations choose* $B = \lceil\sqrt{T}\rceil$*. Then (4.23) is* $O(T^{-1/2})$ *at* $T\lceil\sqrt{T}\rceil$ *sample evaluations. This is not a single-sample* $O(T^{-1/2})$ *theorem.*

Proof. Take expectations of the finite pathwise inequality; no limit interchange is needed for (4.23). If the error-variance series is finite, the expected partial sums of $\|\xi_t\|^2$ are uniformly bounded. Monotone convergence shows that their infinite sum has finite expectation and is therefore finite almost surely. The pathwise theorem then bounds the infinite gradient-energy series almost surely, implying pointwise convergence to zero as in Corollary 4.10. Taking expectations in (4.19) bounds the series of expected squared gradients, which likewise has terms tending to zero. For the batch assertion, expanding the squared norm of the average noise gives B diagonal contributions at most σ^2 and zero cross terms by conditional independence and zero means. Division by B^2 gives $\sigma_t^2 \le \sigma^2/B$, and substitution proves the rate and sample count. □

Corollary 4.12 (Relative oracle error)**.** *Suppose instead that* $\mathbb{E}[\|\xi_t\|^2 \mid \mathcal{F}_{t-1}] \le r^2\,\|h_t\|^2$ *and* $\vartheta = 13C_\beta(M/a)^2r^2 < 1$*, with all assumptions of Theorem 4.9 holding almost surely. Then*

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\,\|h_t\|^2 \le \frac{9\Delta}{(1-\vartheta)\alpha a T},$$

and gradients converge to zero in mean square and almost surely.

Proof. Take expectations in (4.19) and apply the conditional error bound. Move $\vartheta\sum_t \mathbb{E}\,\|h_t\|^2$ to the left and divide by $1-\vartheta > 0$. The resulting bound on all finite partial sums implies summability of expected squared gradients. Monotone convergence implies almost-sure summability too, and the same nonnegative-series argument proves both assertions. Required finite moments follow, for each finite horizon, from $\|g_t\| \le G$, $\|\widehat{m}_t\| \le G$, and $\|x_t\| \le \|x_1\| + (t-1)\alpha MG$, together with Lipschitz continuity of the gradient. □

Remark 4.13 (What persistent noise permits here). For a fixed nonzero variance bound, (4.23) has a nonvanishing term $13C_\beta(M/a)^2\sigma^2$. Taking a smaller constant step does not remove this term. This is an upper bound, not a proof of a matching error floor, but it cannot be cited as convergence to stationarity. The very small relative-error condition is a sufficient hypothesis, not a property implied by unbiasedness or the implementation.

4.7 Stronger and specialized results

Proposition 4.14 (Sharper no-momentum deterministic bound)**.** *Under Assumptions 4.1–4.2, with* $\beta_1 = 0$ *and* $g_t = h_t$*, any* $0 < \alpha \le a/(LM^2)$ *gives*

$$f(x_{t+1}) \le f(x_t) - \frac{\alpha a}{2}\,\|h_t\|^2, \qquad \frac{1}{T}\sum_{t=1}^{T}\|h_t\|^2 \le \frac{2\Delta}{\alpha a T}.$$

If, along the trajectory, $\|\nabla f(x)\|^2 \ge 2\mu(f(x) - f_*)$ *for* $\mu > 0$ *and additionally* $\alpha a\mu \le 1$*, then* $f(x_{T+1}) - f_* \le (1-\alpha a\mu)^T\Delta$*.*

Proof. Now $\widehat{m}_t = h_t$. Smoothness and the scaling bounds give $f(x_{t+1}) \le f(x_t) - \alpha a \|h_t\|^2 + (L\alpha^2 M^2/2)\|h_t\|^2$. The step condition proves the one-step claim, and summing proves the average bound. Inserting the stated gradient dominance inequality gives a nonnegative contraction factor $1 - \alpha a \mu$ at each step; iterating that inequality T times proves the last assertion. □

The gradient dominance condition holds for a differentiable μ-strongly convex function with attained minimum, taking $f_* = f(x_*) = \min f$: strong convexity gives $f_* \ge f(x) + \langle \nabla f(x), x_* - x\rangle + (\mu/2)\|x_* - x\|^2 \ge f(x) - \|\nabla f(x)\|^2/(2\mu)$ by completing the square. The bounded-gradient hypothesis here is along the trajectory, so it does not assert globally bounded gradients for a strongly convex function.

Proposition 4.15 (A convex consequence of the full-momentum theorem)**.** *Under Corollary 4.10, suppose f is convex, has minimizer x_*, $f_* = f(x_*)$, and $\|x_t - x_*\| \le R$ for all t. For $\bar{x}_T = T^{-1}\sum_{t=1}^{T} x_t$,*

$$f(\bar{x}_T) - f_* \le 3R\sqrt{\frac{\Delta}{\alpha a T}}.$$

Proof. Convexity gives $f(x_t) - f_* \le \langle h_t, x_t - x_*\rangle \le R\|h_t\|$. Jensen's inequality for the convex function f and Cauchy–Schwarz for the finite sum give $f(\bar{x}_T) - f_* \le RT^{-1}\sum_t \|h_t\| \le R\sqrt{T^{-1}\sum_t \|h_t\|^2}$. Apply Corollary 4.10 and take the square root. The bounded-distance assumption is explicit; it was not derived from smoothness. □

Proposition 4.16 (Single-coordinate adaptive-step bound)**.** *For the Adam adaptive term in (4.8), with $\beta_2 > \beta_1^2$, $\beta_1, \beta_2 \in [0,1)$, $\epsilon > 0$, $\eta_t > 0$, zero initial moments, $0 \le u_t \le 1$, and arbitrary finite gradients, its magnitude divided by η_t in each coordinate is at most*

$$\frac{(1-\beta_1)\sqrt{1-\beta_2^t}}{(1-\beta_1^t)\sqrt{1-\beta_2}}\sqrt{\sum_{j=0}^{t-1}(\beta_1^2/\beta_2)^j} \le \frac{1}{\sqrt{(1-\beta_2)(1-\beta_1^2/\beta_2)}}. \tag{4.24}$$

This bounds the adaptive step, not the decay contributions or generalization error, and does not prove an advantage over AdamW.

Proof. Cauchy–Schwarz applied to $\sum_{j=0}^{t-1}\beta_1^j g_{t-j,i} = \sum_{j=0}^{t-1}(\beta_1/\sqrt{\beta_2})^j(\sqrt{\beta_2})^j g_{t-j,i}$ gives

$$m_{t,i}^2 \le \frac{(1-\beta_1)^2}{1-\beta_2}\left(\sum_{j=0}^{t-1}(\beta_1^2/\beta_2)^j\right) v_{t,i}.$$

If $v_{t,i} = 0$, every corresponding gradient and $m_{t,i}$ are zero. Otherwise divide by $(\sqrt{v_{t,i}} + \epsilon)^2 \ge v_{t,i}$, multiply by $u_t^2(1-\beta_2^t)/(1-\beta_1^t)^2$, and take square roots. The geometric sum is at most $1/(1-\beta_1^2/\beta_2)$; $(1-\beta_1)/(1-\beta_1^t) \le 1$ and $\sqrt{1-\beta_2^t} \le 1$ give the second bound. No probability statement is involved. □

Sensitivity and limits of the constants. The cancellation cap enters through c; stronger allowed cancellation reduces a and worsens these worst-case constants. This is not evidence that actual trajectories worsen. EMA coefficients .9 and .99 determine responsiveness and the bound 100; the main theorem only needs the resulting trust interval. For a constant nonzero gradient norm R, $A_k = R$ and $Z_k = R/(R+\epsilon)$ for $k \ge 2$, so F_k, S_k tend to this nonzero value. Thus small absolute norms do not generally imply a small normalized detector before the epsilon scale is reached. Lemma 4.4 supplies the correct asymptotic assertion. There is no online-regret theorem or proven improvement in training efficiency in this work.

For scale, setting $L = G = 1$, $\beta_1 = .9$, $\beta_2 = .999$, $\epsilon = 10^{-8}$ and $c = .95$ gives a sufficient step upper bound of about 2.64×10^{-18} in (4.18). This worst-case estimate does not certify an ordinary training step such as 10^{-3}. Its conservatism is a substantial practical limitation, not a claimed necessary stability threshold.

4.8 Counterexamples and limitations

Proposition 4.17 (A fixed-step two-cycle in a legal configuration)**.** *Let $f(x) = x^2/2$, $\beta_1 = \beta_2 = 0$, $\gamma = \lambda = q = 0$, no preprocessing, and $\epsilon > 0$. For constant $\alpha > 2\epsilon$ and $x_1 = \alpha/2 - \epsilon > 0$, the real recurrence has $x_{t+1} = -x_t$. It does not converge to stationarity.*

Proof. The exact step is $x^+ = x - \alpha x/(|x| + \epsilon)$. At $|x| = \alpha/2 - \epsilon$ the denominator is $\alpha/2$, so $x^+ = -x$. Induction gives the cycle, with constant nonzero squared gradient $(\alpha/2 - \epsilon)^2$. This is a smooth strongly convex lower-bounded objective and a legitimate configuration. It does not contradict the small step requirement in Theorem 4.9. □

Proposition 4.18 (Current-sample normalization can reverse expected descent)**.** *At $x = 0$ for $f(x) = (x + .1)^2/2$, use an unbiased stochastic gradient equal to 10 with probability .1 and -1 with probability .9. In the first Adam step with $\beta_1 = \beta_2 = 0$, $\epsilon = .01$, and $\lambda = q = 0$, the expected update direction points uphill, both with inactive chaos and with the active first-step detector using fixed admissible settings. In the latter case let its common trust be $u > 0$. Then every $\alpha > 0$ increases expected objective in this first step.*

Proof. The gradient mean is $1 - .9 = .1 = f'(0)$; the oracle can be extended to every x as $g(x) = x + Z$ with the displayed two-point Z, and its noise has finite constant variance. At the first step $F = S = 1$ and $D = 0$, so the detector trust is independent of the gradient magnitude and has the same value u in both outcomes. The mean normalized direction is

$$\mathbb{E}\left[\frac{ug}{|g| + .01}\right] = u\left(\frac{100}{1001} - \frac{90}{101}\right) < 0.$$

Writing $V = ug/(|g| + .01)$, direct expansion of the quadratic gives $\mathbb{E}f(-\alpha V) - f(0) = -.1\alpha\mathbb{E}V + (\alpha^2/2)\mathbb{E}V^2 > 0$. This disproves a universal one-step expected-descent inequality under unbiasedness alone. A single uphill step is not a proof of asymptotic nonconvergence, and we make no such inference from this example. □

Proposition 4.19 (Bounded positive cancellation alone is insufficient)**.** *For the same two-point gradient, a current-gradient multiplier in $[.01, 1]$ can have $\mathbb{E}[ug] < 0$ despite $\mathbb{E}g > 0$.*

Proof. Take $u(10) = .01$ and $u(-1) = 1$. Then $\mathbb{E}[ug] = .01 - .9 = -.89$ while $\mathbb{E}g = .1$. This generic multiplier example rules out a proof based only on its interval bounds. It is not asserted to be the exact implemented detector; a reachable-history calculation for that detector is recorded separately in the following proposition and `counterexamples.md`. □

Proposition 4.20 (Anti-alignment from the exact detector)**.** *Set $\beta_1 = \beta_2 = 0$, $\epsilon = .01$, $\gamma = p_{\text{ext}} = 1$, $c_{\max} = 1$, zero warmup and horizon, adaptive thresholding with $\tau_{\text{scale}} = 2$, auto-gamma off, and no decay or preprocessing. There is a reachable history and an unbiased next gradient with positive conditional mean for which the implemented trust satisfies $\mathbb{E}[u_2 g_2 \mid \mathcal{F}_1] < -463/2000$.*

Proof. After a first gradient -1, the old momentum is -1 and $A_1 = F_1 = S_1 = 1$. At step two take $g = 10$ with probability $1/10$ and $g = -1$ otherwise. Its conditional mean is $1/10$. For $g = 10$, direct substitution gives

$$A_+ = 109/100, \quad Z_+ = 100/11, \quad F_+ = 199/110,$$
$$S_+ = 1189/1100, \quad F_+/(S_+ + \epsilon) = 199/120, \quad D_+ = 2001/2002 > 99/100.$$

Thus the sigmoid excess exceeds $1/2$, and $s_+ > (3/4)(1/2)(1/2 + 99/100) = 447/800$, since $S_+ > 1$ and $\tanh(1) > 3/4$. The latter inequality follows from $\exp(1) > 1 + 1 + 1/2 + 1/6 = 8/3$ and therefore $\exp(2) > 7$. The cap .99 also exceeds $447/800$, so $u_+ < 353/800$. For $g = -1$ the values are

$$A_- = 1, \quad F_- = 1009/1010, \quad S_- = 10099/10100,$$
$$F_-/(S_- + \epsilon) = 1009/1020 < 1, \quad D_- = 1/202 < 1/200.$$

The excess is less than $1/2$ and $\tanh S_- < 1$; hence $s_- < 101/400$ and $u_- > 299/400$. It follows that

$$\mathbb{E}[ug \mid \mathcal{F}_1] = u_+ - (9/10)u_- < 353/800 - (9/10)(299/400) = -463/2000.$$

To realize this history with an everywhere unbiased oracle for a smooth lower-bounded objective, fix $\alpha > 0$ and start at zero. Let $u_1 > 0$ be its deterministic first-step trust and $\delta = \alpha u_1/(1+\epsilon) > 0$. Set $L_0 = 11/(10\delta)$ and $f(x) = L_0(x - 1/L_0)^2/2$. Then $f'(0) = -1$ and $f'(\delta) = 1/10$. Use the exact gradient except at δ, where the displayed two-point oracle is used. This oracle is unbiased everywhere, has variance at most $1089/100$, and reaches the required history with probability one. Its definition need not be continuous in x; no such continuity was assumed for an unbiased stochastic oracle. This is a finite-step obstruction, not an asymptotic nonconvergence claim. □

Proposition 4.21 (Gradient centralization can stall permanently)**.** *For a 1×2 parameter tensor X, let $f(X) = \|X - (1,1)\|^2/2$ and $X_1 = (0,0)$. With gradient centralization on and initialized zero moments, the implemented real recurrence stays at zero for any learning-rate schedule, in either Adam or Lion mode, even with AGC and either decay enabled, provided its arithmetic is well-defined. The true squared gradient remains 2.*

Proof. At zero the raw gradient is $(-1,-1)$. Whole-tensor AGC multiplies both coordinates by the same scalar, and centralization subtracts their common mean, giving $g = 0$. Both moment states remain zero and the Adam/Lion adaptive update is zero. Both decay operations multiply the zero parameter, preserving it. Induction repeats the argument at every step. Meanwhile $\nabla f(0) = (-1,-1)$, of squared norm 2. Thus projected-gradient reasoning cannot silently be used to claim stationarity of the original objective. □

AGC, decay, and accepted but invalid inputs. Clipping the two-point oracle above to magnitude one gives mean $.1 - .9 = -.8$, reversing its original mean. Whole-tensor AGC realizes this at a parameter norm and AGC coefficient whose threshold in (4.1) equals one. Unbiasedness must therefore be imposed on the *postprocessed* oracle or its bias must be controlled explicitly. Under the standard unclipped oracle assumption this is not automatic.

For $f(x) = (x-1)^2/2$ at $x = 1$ with zero initial moments, weight decay $\lambda > 0$ and $\alpha\lambda \neq 0$ moves a stationary point to $1 - \alpha\lambda$. In general the no-quantum-decay direction is $H_t\widehat{m}_t + \lambda x_t$, not $H_t(\widehat{m}_t + \lambda x_t)$. When H_t varies with time and history, this is not the gradient step for a fixed objective $f + \lambda\|x\|^2/2$. Quantum decay further adds a current-gradient-dependent coordinate shrinkage, with the cross term in (4.8). Neither is covered by the main theorem. A conditional theorem for a specially chosen fixed metric would not establish the default case.

Negative p_{ext}, which is not rejected by the constructor, can produce $u > 1$ because the clamp has only an upper bound. Accepted $\epsilon = 0$ with zero gradient and variance makes the Adam quotient $0/0$; nonfinite hyperparameters or data cause further failures. Large $\eta\lambda$ or ηQ_k can change the sign of a decay factor. Such decay is outside the main theorem's zero-decay configuration. Invalid epsilon, negative p_{ext}, and nonfinite inputs are outside the analytic domain; large positive decay products can be legal finite inputs, but the decay factors need not be contractions.

Finite precision. Even finite positive epsilon and bounded trust do not prevent floating-point stalling, overflow, underflow, or cancellation. If a nonzero real increment is smaller than half an ulp at a stored parameter and rounding returns the old value, the real update and stored trajectory differ. No summable rounding error bound is established for this implementation. The real-arithmetic stationarity theorem therefore does not imply exact convergence of fp16, bfloat16, fp32, or fp64 program executions.

4.9 Relationship to adaptive optimization theory

Adam introduced adaptive first and second moments with bias correction [1]. Its original convergence arguments cannot be used as an unconditional inheritance theorem: Reddi et al. [5] exhibit failure and motivate AMSGrad's long-term second-moment memory. PsiLogic's current variance has no running maximum, so it is not AMSGrad. Adding one would be an algorithmic modification and would still require an analysis of the extra current-gradient multiplier.

Chen et al. [6] give sufficient conditions for classes of Adam-type methods and analyze effective-step behavior. We do not assert those conditions are implied by this detector. Theorem 4.9 instead controls the momentum tracking error pathwise under a uniformly positive bounded scaling and a small constant step. Its deterministic and vanishing-error assumptions explain why it is compatible with known stochastic failures.

Decoupled weight decay differs from loss-based L_2 regularization for adaptive methods [3]; hence our stationarity target and decay exclusions are essential. The Lion branch follows a different sign-momentum structure [4] and is specified but not covered by the Adam proof. The simplified stochastic benchmark connects to smooth nonconvex stationarity bounds [7]. Matching its iteration rate under stronger conditions or larger batches is not evidence of improved oracle complexity, robustness, or training efficiency.

4.10 Implementation-to-mathematics correspondence

The authoritative import/package root is `psilogic/`. Public constructors, presets, convenience selection, integrations, and group builders ultimately construct `PsiLogic`; they do not introduce another optimizer update implementation. The generated `build/lib/psilogic` copy has stale CUDA files and is separately recorded, not silently substituted for the active source. Untracked paper scripts do not alter the update path. The full trace and defaults are in `implementation_trace.md` and `algorithm_specification.md`; detailed kernel scope is in `backend_trace.md`.

Mathematical object	Source location in inspected working tree
AGC, centralization, initial state	`optimizer.py:96--126`
Defaults, parameter options	`optimizer.py:283--344`
Schedules and warmup	`_chaos.py:35--61,114--124`
Disagreement and soft signal	`_chaos.py:127--186`
Trust and hard cap	`_chaos.py:186--207`
Norm EMA initialization/update	`_chaos.py:263--282`
Sequential decay, active branching	`optimizer.py:472--527`
Adam/Lion update and epsilon	`optimizer.py:529--558`
Scalar preparation and old momentum	`optimizer.py:560--678`
Foreach preparation/finalization	`optimizer.py:680--917`
Dispatch	`optimizer.py:920--974`
Fused orchestration	`_cuda/step.py:320--668`
Fused Adam/Lion arithmetic	`_cuda/kernels.py:239--373`

Paths in the table are relative to `psilogic/`. References denote the snapshot, not a promise that future edits retain line numbers.

The scalar branch is used on CPU and when compiling; ordinary CUDA dispatch prefers fused kernels, then foreach. On finite well-defined inputs, the intended real recurrences agree for supported dense contiguous CUDA layouts with fresh gradients each call. This statement is deliberately narrower than bitwise equivalence: moments are normally fp32; low-precision norm arithmetic, fused reductions, host versus tensor auto-gamma, and intermediate parameter rounding differ. Fused quantum decay evaluates a sigmoid identity for tanh. Fused flat pointer indexing does not validate parameter contiguity; mixed CPU/CUDA groups can reach an unsuitable launch. Foreach and homogeneous fused AGC mutate gradient buffers in place, unlike

the scalar path. Repeated steps without replacing gradients can therefore differ even in the real model. DDP averages fast/slow only and does not guarantee identical trust for ranks with different local gradients or momentum. Resumed state may have different initial values or dtypes and needs separate hypotheses.

4.11 Formal verification status

Ten declarations in `lean/PsiLogic.lean` were successfully compiled and kernel-checked with Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The proof-check command exited with status zero. The raw output, source hash, command, and toolchain are in `lean/build.log` and `lean/build_status.json`.

The checked statements cover the cap-dependent trust interval and .01 floor; the normalized-EMA upper bound and preservation of the interval $[0, 100]$; scalar identities for epsilon bias correction and denominator scaling; absorption of a scalar smoothness-error term; finite telescoping; a weighted certificate from a terminal objective lower bound; and the corresponding finite average $1/T$ certificate. The last four statements assume the relevant scalar descent inequality explicitly. They do not derive it from differentiability, the vector optimizer, or a stochastic process. All ten axiom reports list only `propext`, `Classical.choice`, and `Quot.sound`; there are no placeholder proofs or added axioms.

This is meaningful verification of a limited algebraic core, not a formal proof of Theorem 4.9, the stochastic corollaries, Python execution semantics, or Triton memory-layout correctness. Remaining formalization work includes the normed-space smoothness argument, vector momentum convolution bounds, the connection to the actual recurrence, and conditional-expectation and infinite-series results. Neither numerical parity nor repeated mathematical review substitutes for these missing formal developments.

4.12 Reproducibility, audit, and theorem status

The checks in this section describe the original investigation snapshot. AI agents prepared the source inspection, proof audit, and second derivation; these checks are not external peer review. Later publication edits to comments and documentation are recorded separately from the original source hashes.

Before proof construction, an independent double-precision recurrence was compared with actual CPU scalar and explicitly invoked CPU foreach execution. The check passed 34 cases and 47,380 parameter/state comparisons at absolute and relative tolerance 2×10^{-5}, including Adam, Lion, AGC followed by centralization, quantum decay, warmup, cosine schedules, auto-gamma, missing gradients, unequal counters, zero-norm initialization, and the epsilon placement. These are numerical correspondence tests, not proofs of exact arithmetic equivalence. GPU execution was unavailable. The source hashes and results are retained with the scripts.

The selected existing regression suite passed 72 tests, with 16 CUDA-dependent tests skipped. Separate checks verified exact symbolic identities and tested 300 finite momentum-tracking sequences and eight synthetic trajectories against the proof inequalities. Separate checks reproduced the analytic counterexamples. At the time of the investigation, SHA-256 comparisons confirmed that all 14 inspected package source files were unchanged. Build commands and results accompany the artifacts.

The proof audit tracks conditional measurability, initial-state terms, coefficient signs, index shifts, and the distinction between pathwise and expected statements. In particular: the current detector is never treated as predictable; the first oracle error is not dropped; the momentum drift sum ends at $T - 1$ before being enlarged; the step bound is used before error absorption; and only summable nonnegative series justify last-iterate conclusions. A second derivation and algebraic stress checks are recorded separately; neither is represented as formal verification.

Result	Scope/status of mathematical claim
Detector/EMA and trust bounds	Complete real-arithmetic proofs; explicit positive-domain assumptions
Predictable stochastic $O(T^{-1/2})$	Complete proof for modified simplified method only
Full-momentum bound (4.19)	Complete pathwise proof for Assumption 4.2
Deterministic $O(T^{-1})$ and stationarity	Consequences under bounded gradients and small constant step
Stochastic vanishing/relative error	Conditional consequences; batch/sample cost stated
No-momentum gradient dominance	Conditional linear objective convergence
Counterexamples	Exact analytic constructions; one-step and asymptotic claims distinguished
Default decay, arbitrary fixed noise, Lion	No general stationarity theorem established
Floating point, GPU/DDP semantics	Not formally verified; source-level limitations disclosed
Lean scalar and finite-sum core	Ten declarations kernel-checked; full convergence theorem not formalized

Remaining mathematical limitations are substantive. There is no theorem for the full default configuration with positive weight decay, arbitrary persistent unbiased noise, centralization, quantum decay, or general Lion updates; no guarantee for arbitrary group clocks or loaded checkpoints; no proof that practically used learning rates meet the conservative sufficient bound; and no theorem of superiority over AdamW. These exclusions are visible assumptions and open obligations, not hidden gaps in a claimed broader result. The separate proof-gap register distinguishes rejected claims from proved conditional results. The original investigation left the optimizer source and pre-existing research artifacts unchanged. It did not publish or submit them.

5 Evidence and protocol

5.1 Snapshots and provenance

The reference is `benchmark/results/full`; the later partial snapshot is `benchmark/results/fused_h100`. Means and sample SD are recalculated from summaries and checked against aggregate CSVs. Each reported combination has successful runs for seeds 0, 1, and 2. The later summary has only NLP, ViT, and ResNet, although its configuration requests diffusion. A requested task without outcomes is not a completed experiment.

Both configs specify H100 80GB HBM3, batch 64, BF16 AMP, global clipping at 1, cosine LR with 100 warmup steps, seven log-spaced LR candidates from 1e-5 to 1e-2, sweep limits of 500 steps or 5 epochs, and evaluation limits of 2000 steps or 10 epochs. The loop stops at the applicable step or epoch limit. These are configured budgets: absent later step histories, we cannot confirm every later run completed 2000 updates.

Arena definitions describe a small GPT on TinyStories, ViT on CIFAR-100, ResNet-18 on Tiny ImageNet, and `DDPM/UNet` diffusion on CelebA. Metadata does not fully serialize historical architecture, transforms, or resolved optimizer arguments. ViT also has a package-dependent model fallback; future runs should record the actual architecture.

The reference environment is PyTorch 2.4.1+`cu124/CUDA` 12.4; the later environment is 2.8.0+`cu128/CUDA` 12.8. Later ViT peak allocation is about 5497 MiB for PsiLogic, compared with roughly 1229 MiB in the reference. These changes and different selected rates prevent a controlled backend-only comparison. The later directory name indicates a fusion evaluation, but the config lacks per-run backend records. We report outcomes, not a verified kernel trace.

5.2 Statistics

For this audit, we select one primary outcome per arena: NLP perplexity, vision accuracy, and diffusion MSE. NLP learning-rate selection in the harness uses validation loss; exponentiating loss preserves outcome ordering but does not preserve the paired t-test statistic. Two-sided seed-paired tests compare the audit's selected outcomes. Holm correction covers all arena-by-baseline primary comparisons within each snapshot: 12 reference tests and 9 later tests. This family differs from the reference's saved unpaired Welch tests and the later saved analyzer's per-arena, multiple-metric family. Original significance files remain unchanged; new results are written separately.

Tables give raw and adjusted p-values. Three pairs leave two degrees of freedom, and normality of differences cannot be meaningfully assessed. This retrospective analysis was not preregistered. Failure to reject does not establish equivalence; significance does not establish usefulness. Pairing assumes matching seed labels represent matched conditions, as intended by the harness. CSVs alone cannot verify initialization or data order.

5.3 Timing definition

Saved final wall time covers the training loop, periodic evaluation, and final evaluation. It excludes preparation, LR selection, and `model/optimizer` construction. The current harness synchronizes step timing; exact historical behavior needs the missing source revisions. Ratios divide mean run times, not optimizer-only times or individual seed ratios. The optional optimizer host profiler is asynchronous on CUDA and cannot replace synchronized GPU measurements. No time-to-target-quality claim is made.

6 Later partial results

6.1 Quality

Table 1. Later primary metrics, mean ± sample SD over three seeds. Accuracy is a fraction. Bold marks the best observed mean, not significance.

Primary metric	Adam	AdamW	Lion	PsiLogic
NLP perplexity (lower)	13.81 ± 0.06	8.20 ± 0.20	20.50 ± 1.29	**8.15 ± 0.16**
ViT accuracy (higher)	0.1169 ± 0.0030	0.2825 ± 0.0047	**0.3049 ± 0.0036**	0.2672 ± 0.0067
ResNet accuracy (higher)	0.1710 ± 0.0042	**0.2214 ± 0.0052**	0.2075 ± 0.0049	0.2208 ± 0.0023

Table 2. Validation losses from the same later runs, descriptive secondary outcomes.

Validation loss	Adam	AdamW	Lion	PsiLogic
NLP	2.626 ± 0.004	2.104 ± 0.024	3.019 ± 0.064	2.098 ± 0.020
VIT	3.880 ± 0.002	2.886 ± 0.013	2.764 ± 0.011	2.973 ± 0.017
RESNET	3.813 ± 0.009	3.456 ± 0.009	3.509 ± 0.023	3.467 ± 0.008

The NLP mean is about 0.6% below AdamW, much smaller than the reference gap of about 4.7%; the later paired test does not detect an advantage. ViT trails AdamW by about 1.54 percentage points and Lion by about 3.77. ResNet differs from AdamW by about -0.06 points. After the nine-test Holm correction, the ViT loss against AdamW remains detectable (adjusted $p = 0.0294$), whereas NLP and ResNet differences do not. The later timing results cannot be presented as preserving the earlier ViT advantage.

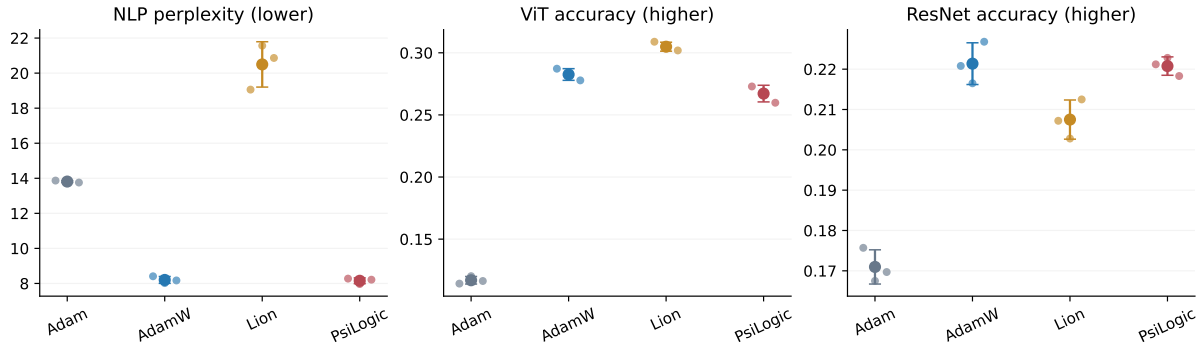


Figure 1: Later partial snapshot: individual seed points and mean with sample SD for all four optimizers. Accuracy is a fraction. No later diffusion outcomes or learning curves are available.

6.2 Wall time and memory

Table 3. Later run wall time in seconds, mean ± sample SD; final column is ratio of means.

Arena	Adam	AdamW	Lion	PsiLogic	`Psi/AdamW`
NLP	20.49 ± 0.17	20.22 ± 0.07	19.90 ± 0.07	22.22 ± 0.18	1.099×
VIT	113.58 ± 1.90	114.45 ± 3.76	112.69 ± 1.88	117.18 ± 0.34	1.024×
RESNET	36.08 ± 0.49	35.87 ± 0.38	35.34 ± 0.18	37.25 ± 0.72	1.038×

Overhead relative to AdamW is about 9.9% for NLP, 2.4% for ViT, and 3.8% for ResNet. This supports modest end-to-end overhead in these recorded workloads. It does not show PsiLogic is faster than AdamW, quantify optimizer-only overhead, or isolate fusion's contribution.

Table 4. Peak allocated GPU memory in MiB (bytes divided by 1024 squared), mean ± sample SD. This is not total board usage or reserved memory.

Peak allocated MiB	**Adam**	**AdamW**	**Lion**	**PsiLogic**
NLP	458.4 ± 0.1	458.4 ± 0.1	446.0 ± 0.1	458.6 ± 0.1
VIT	5491.9 ± 5.4	5492.6 ± 3.3	5164.4 ± 5.3	5497.2 ± 4.2
RESNET	857.3 ± 0.4	855.6 ± 0.4	812.7 ± 0.4	857.8 ± 0.4

Within later arenas, PsiLogic and AdamW have similar allocations. Lion uses less memory. There is no demonstrated memory advantage for PsiLogic; the large cross-snapshot ViT memory change is unexplained by saved metadata.

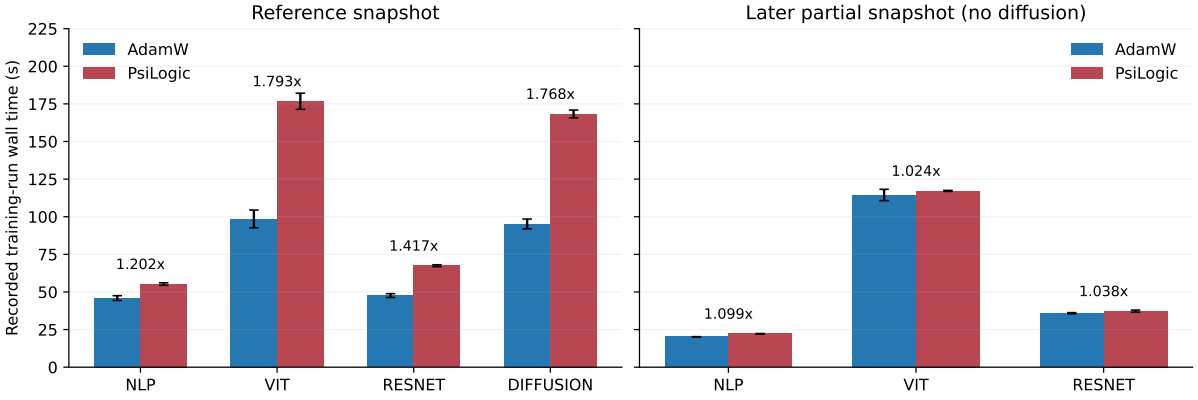


Figure 2: Recorded wall time for AdamW and PsiLogic by snapshot. Bars show means, error bars sample SD, and labels ratios of means. Environments and some selected rates differ; no later diffusion measurement exists.

6.3 Paired comparisons and rates

The saved optimizer-only profile gives a different cost picture. Its CUDA workload has 202,506 parameters in 50 tensors. The accompanying profiling script compares against foreach AdamW. The saved file reports one median per method, without repeated-run uncertainty or raw timing samples:

Method	**Saved median step (ms)**	**Ratio to AdamW**
AdamW	0.603	1.00×
PsiLogic (foreach)	4.986	8.27×
PsiLogic (fused)	1.371	2.27×

Fusion is much faster than PsiLogic's foreach path on this small workload, but the fused step is still about 2.27 times the AdamW step. That is compatible with a small full-training overhead when `forward/backward` work dominates, although this separate microbenchmark does not establish the optimizer's share of the three arena runtimes. The saved run note is dated 24 September 2026. Its availability check for Triton supports the intended fused evaluation, but does not replace per-run backend traces or provide a matched comparison with the older environment.

Table 5. Later primary-quality tests, raw p / Holm-adjusted p, across nine comparisons. Direction follows Table 1.

Arena	vs Adam raw / Holm	vs AdamW raw / Holm	vs Lion raw / Holm
NLP	0.000275 / 0.00248	0.443 / 0.886	0.00446 / 0.0235
VIT	0.000882 / 0.00706	0.00735 / 0.0294	0.00338 / 0.0235
RESNET	0.00336 / 0.0235	0.902 / 0.902	0.0843 / 0.253

Table 6. Later selected base learning rates.

Arena	Adam	AdamW	Lion	PsiLogic
NLP	0.000316	0.000316	0.000316	0.000316
VIT	3.16e-05	3.16e-05	1e-05	3.16e-05
RESNET	0.0001	0.000316	0.0001	0.000316

7 Historical reference outcomes

These four-arena results are the reference experiment reported in the original arXiv preprint [2]. They are retained here with recalculated statistics, separately from the quality associated with later timing.

Table 7. Reference primary metrics, mean ± sample SD; bold is best mean.

Primary metric	Adam	AdamW	Lion	PsiLogic
NLP perplexity (lower)	13.66 ± 0.22	8.17 ± 0.08	21.04 ± 1.41	**7.79 ± 0.18**
ViT accuracy (higher)	0.0787 ± 0.0035	0.2229 ± 0.0024	0.2127 ± 0.0018	**0.2444 ± 0.0060**
ResNet accuracy (higher)	0.1716 ± 0.0044	0.2192 ± 0.0052	0.2045 ± 0.0070	**0.2221 ± 0.0014**
Diffusion MSE (lower)	**0.01987 ± 0.00006**	0.01987 ± 0.00006	0.02175 ± 0.00025	0.02009 ± 0.00045

Table 8. Recalculated reference tests, raw p / adjusted p across 12 comparisons.

Arena	vs Adam raw / Holm	vs AdamW raw / Holm	vs Lion raw / Holm
NLP	0.000823 / 0.00906	0.0255 / 0.153	0.00394 / 0.0394
VIT	0.000518 / 0.00622	0.0465 / 0.232	0.0174 / 0.122
RESNET	0.00439 / 0.0395	0.512 / 1	0.0586 / 0.235
DIFFUSION	0.435 / 1	0.435 / 1	0.00503 / 0.0403

These replace the old unadjusted Welch significance claims. Mean ordering is unchanged; inference must follow the recalculated tests rather than old stars. Reference means favor PsiLogic in NLP and ViT, give a small ResNet difference from AdamW, and slightly worse diffusion MSE than `Adam/AdamW.` None of the four primary comparisons against AdamW meets the adjusted 0.05 threshold: adjusted p is 0.153 for NLP, 0.232 for ViT, and 1 for ResNet and diffusion. These data therefore do not establish a corrected quality advantage over AdamW, including on diffusion.

Table 9. Reference run wall time in seconds and ratio of means.

Arena	Adam	AdamW	Lion	PsiLogic	Psi/AdamW
NLP	46.56 ± 1.38	45.94 ± 1.59	38.19 ± 7.36	55.23 ± 0.83	1.202×
VIT	95.18 ± 8.85	98.54 ± 5.89	98.57 ± 5.59	176.73 ± 5.36	1.793×
RESNET	45.26 ± 1.09	47.59 ± 1.27	46.13 ± 1.46	67.44 ± 0.64	1.417×
DIFFUSION	94.21 ± 0.95	95.20 ± 3.27	91.62 ± 1.67	168.32 ± 2.59	1.768×

Reference ratios are about 1.20×, 1.79×, 1.42×, and 1.77× for NLP, ViT, ResNet, and diffusion. They describe that environment, not later partial measurements. The difference documents an observed cost change without attributing all of it to fusion.

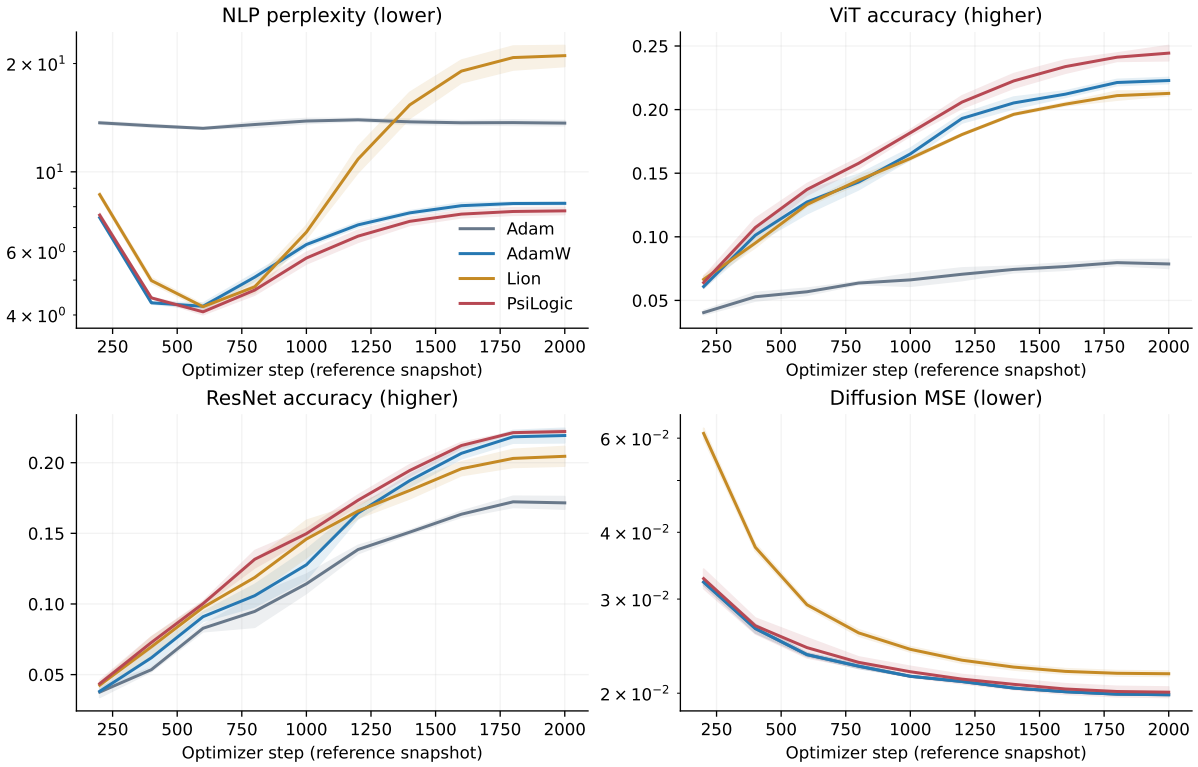


Figure 3: Historical reference validation curves regenerated from steps.csv. Lines show means and bands sample SD across three seeds at recorded steps. NLP and diffusion axes are logarithmic. These are not later curves or new training measurements of current source.

8 Discussion and limitations

The clearest engineering observation is smaller relative full-run overhead in the later snapshot. This coexists with a 2.27-fold optimizer-step cost in the separate saved microbenchmark; the two measurements answer different questions. Quality is less consistent: the older NLP and ViT advantages are not both present later. Software, selected rates, unrecorded `architecture/configuration` details, and possible implementation changes remain candidate explanations. Measurements do not isolate these factors or separate cancellation from preprocessing and presets.

Other exploratory NLP follow-ups and older experiments exist. We do not pool them because their settings differ. This is a retrospective report in a project with multiple experiments, not preregistered confirmation. Unit tests establish software properties under tested conditions, not better generalization.

Evidence is limited to three seeds per combination, small models, and short configured budgets. There is no later diffusion outcome, later step history, matched time-to-quality analysis, or external replication established by these artifacts. The convergence theorem in Section 4 applies under its stated restrictions; it does not certify the finite-precision benchmark configurations or explain their quality ordering. Exact training commits, resolved settings, and backend traces are absent. Validation used for LR selection is not an untouched test set. Equal LR candidate counts do not equalize all search effort.

The detector measures relative norm changes and disagreement, not curvature, noise variance, or proof a step is harmful. Its baseline sigmoid value allows damping under steady gradients. Small gradients do not guarantee zero damping. Secondary decay is separate, and epsilon placement makes the zero-cancellation limit different from standard AdamW.

A controlled follow-up should pin code, architecture, data processing, effective batch, precision, and environment; record resolved arguments and backend; compare execution paths under matching settings; and repeat all four tasks with more seeds. Ablations should separately disable cancellation, preprocessing, and secondary decay. Fixed-step quality and time to a prespecified target should both be reported, including unsuccessful runs. These are proposed experiments, not completed evidence.

9 Reproducing this revision

The empirical source is `PAPER.md`; the theoretical source is `research/convergence/manuscript.tex`. The combined article is assembled by `research/unified/build.py`. Tables, figures, paired tests, and file hashes are derived from saved measurements:

```
python scripts/paper_results.py
python scripts/build_arxiv_pdf.py
python3 research/unified/build.py
```

Analysis requires NumPy, SciPy, and Matplotlib; PDF compilation requires pdflatex or the repository's Tectonic executable. The combined output is `output/pdf/paper_unified.pdf`, with editable LaTeX and a shared bibliography in `research/unified/`. The standalone empirical build still writes `output/pdf/paper.pdf` and its original copies. Analysis and provenance hashes are in `arxiv/generated`. Hashes identify analyzed files, not unknown training revisions. Source CSVs remain unchanged. No training or downloads occur during this analysis.

For scholarly citation of PsiLogic, use the original arXiv preprint [2]: arXiv:2607.16268, https://arxiv.org/pdf/2607.16268, article DOI https://doi.org/10.48550/arXiv.2607.16268. The Zenodo software record [8] is a supplementary reference for code and artifacts. The revised equations and reanalysis in this local manuscript are dated separately from the original arXiv version.

To evaluate the current update, install benchmark dependencies and prepare datasets as documented in `benchmark/README.md`. Record the source revision and resolved arguments before starting:

```
pip install -e ".[benchmark,cuda]"
pip install -r benchmark/requirements.txt
cd benchmark
python -m fairbench --data-root ./data --output-dir results/current_rerun
```

This starts a current-default experiment, not a reconstruction of unspecified historical settings. The same optimizer name alone does not make an older table reproducible.

10 Conclusion

PsiLogic adds bounded tensor-wise damping around an Adam-style direction, with optional preprocessing, secondary decay, and Lion mode. The equations include details that change its mathematics. Later H100 measurements show 2.4-9.9% additional recorded run time relative to AdamW across three tasks, near-AdamW NLP and ResNet means, and worse ViT accuracy. The separate optimizer-only microbenchmark still costs 2.27× AdamW for fused PsiLogic. Older four-task outcomes remain separate. These findings justify controlled evaluation, not a universal quality claim or a claim that historical tables validate every detail of today's code. The full-detector Adam analysis proves deterministic stationarity and conditional stochastic bounds under explicit restrictions; the counterexamples and the conservative sufficient step bound delimit what those results establish. Ten checked Lean lemmas verify a scalar algebraic core, while the complete convergence theorem remains an informal mathematical proof.

A Later per-seed primary outcomes

Accuracy is a fraction; tests use seed values before rounding.

Arena / seed	Adam	AdamW	Lion	PsiLogic
NLP / 0	13.86950	8.40920	19.05928	8.27578
NLP / 1	13.81303	8.02130	21.56638	7.96639
NLP / 2	13.75750	8.17168	20.86497	8.21472
VIT / 0	0.11420	0.28720	0.30890	0.27290
VIT / 1	0.12020	0.28260	0.30390	0.26880
VIT / 2	0.11630	0.27780	0.30190	0.25980
RESNET / 0	0.17570	0.22080	0.20720	0.22120
RESNET / 1	0.16750	0.21650	0.20280	0.22280
RESNET / 2	0.16970	0.22680	0.21250	0.21830

B Reference selected rates

Arena	Adam	AdamW	Lion	PsiLogic
NLP	0.000316	0.000316	0.000316	0.000316
VIT	3.16e-05	0.000316	0.0001	0.000316
RESNET	0.0001	0.000316	0.0001	0.000316
DIFFUSION	0.001	0.001	0.0001	0.001

C Reference per-seed primary outcomes

Arena / seed	Adam	AdamW	Lion	PsiLogic
NLP / 0	13.89225	8.18040	19.46536	7.87066
NLP / 1	13.64020	8.09092	21.46757	7.58325
NLP / 2	13.44957	8.24811	22.18221	7.91506
VIT / 0	0.07750	0.22550	0.21440	0.23760
VIT / 1	0.08260	0.22220	0.21080	0.24680
VIT / 2	0.07590	0.22090	0.21300	0.24880
RESNET / 0	0.17510	0.21940	0.20240	0.22090
RESNET / 1	0.16660	0.21400	0.19890	0.22360
RESNET / 2	0.17300	0.22430	0.21230	0.22170
DIFFUSION / 0	0.01985	0.01985	0.02169	0.02004
DIFFUSION / 1	0.01982	0.01982	0.02153	0.01967
DIFFUSION / 2	0.01993	0.01993	0.02202	0.02057

References

[1] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015. URL https://arxiv.org/abs/1412.6980.

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